Midterm 1

DATA 412/612 Statistical Programming in R

October 14, 2026 75 minutes 100 points

Open `midterm-1.qmd`. Put your name in the subtitle. That file already has a heading and a code chunk for each part. Write your answers there. Code is already set to show in the rendered file. Render to HTML (the default). PDF is fine too. Submit the `.qmd` and the rendered file before the 75 minutes are up.

- You may use your notes, the R help files, and the internet.
- You may not communicate with anyone else.
- Use the tidyverse where a tidyverse verb exists. Question 1 is a function and a pipe; base R is appropriate there.
- `library(tidyverse)` is enough. `mpg` is in ggplot2. `starwars` is in dplyr. `mtcars` is in base R.
- Either pipe is fine: `|>` or `%>%`.
- Partial credit is given for code that runs. If you are stuck, go to the next part.

The suggested times are a guide. You have 75 minutes.

1. **A function and a pipe** (20 points, about 14 minutes)

   a. (14 points) Write a function `finite_ratio(x, y)`.

   - If `x` or `y` is not numeric, or the two lengths differ, stop with the message `x and y must be numeric and the same length`.
   - Drop every position where `x` or `y` is not finite. `NA`, `NaN`, `Inf`, and `-Inf` are not finite.
   - If fewer than 2 positions remain, return `NA_real_`.
   - If the mean of the remaining `y` is 0, return `Inf`.
   - Otherwise return the mean of the remaining `x` divided by the mean of the remaining `y`. Use a pipe for that ratio.

   Run it on these four inputs. The chunk in `midterm-1.qmd` already has `#| error: true`, so the last call can stop and the document still renders. Leave that error visible.

```
finite_ratio(c(1, NA, 4, Inf), c(2, 3, 6, 1))
finite_ratio(c(1, 2), c(0, 0))
finite_ratio(c(1, NA_real_), c(NA_real_, 4))
finite_ratio(c(1, 2), "a")
```

   b. (6 points) Using one pipe and no intermediate objects, start from `c(2, 4, 8, NA, 16)`, drop the missing value, take the natural log, take the mean, then exp. Report the number. In one sentence, say what goes wrong if the `NA` is still in the vector when you call `log()`.

2. **Rows, columns, and groups** (20 points, about 15 minutes)

   Use `mpg`. Answer each part with one pipeline.

   a. (8 points) Keep cars from `year` 2008 in class `compact` or `midsize` with `hwy` at least 30. How many cars is that? Show the five with the highest `hwy`. Break remaining ties by `cty`, highest first. Keep only `manufacturer`, `model`, `year`, `cyl`, `cty`, and `hwy`.

   b. (7 points) For each `class`, report the number of cars, the mean of `hwy`, and the median of `cty`. Arrange by mean highway mileage, highest first.

c. (5 points) Add a column `economy` equal to `(cty + hwy) / 2`. Place it immediately after `model`. Drop `trans` and `fl`. Show the column names of the result, in order.

3. **`case_when`, `across`, and distinct rows** (20 points, about 16 minutes)

   a. (8 points) Add a `size` column to `starwars` with `case_when()`, in this order:

   - `"missing"` if `height` is missing
   - `"large"` if `height` is at least 200 or `mass` is at least 200
   - `"droid"` if `species` is `"Droid"`
   - `"other"` otherwise

   Report how many characters fall in each size, largest count first. In one sentence, say what would change about the droid count if `"large"` were not checked before `"droid"`. Name the character.

   b. (7 points) Drop characters whose `species` is missing. Keep species that appear more than once. For each of those species, use `across()` for the median `height` and the median `mass` (ignore missing values inside the medians), and report `n()` as well. `n()` counts every row in the species, including rows whose height or mass is missing. Arrange by `n()`, largest first.

   c. (5 points) Two short pipelines.

   - Drop `starwars` rows where `hair_color` or `skin_color` is missing, then report how many distinct pairs of those two columns remain.
   - `mtcars` stores the car name as a row name. Make that a column called `car`. Keep cars with `mpg` of at least 30. Return `car`, `mpg`, and `cyl`, arranged by `mpg`, highest first.

4. **One picture** (20 points, about 12 minutes)

   From `mpg`, plot highway mileage against engine displacement.

   - Points: map `class` to color inside `geom_point()` only, and set an alpha so overlapping points stay visible.
   - One straight smooth in black, dashed, with no gray band. The smooth is not colored by `class`.
   - Facet by `drv`.
   - Use `theme_bw()`. Label the axes `Displacement (litres)` and `Highway mpg`.

   In two sentences, say what happens to highway mileage as displacement increases, and whether that pattern shows up in every `drv` panel. Name one difference you can see between panels.

5. **A small table, then `across()`** (20 points, about 8 minutes)

   `halls` is already in `midterm-1.qmd`. Do not read a file. There is no CSV.

```
halls <- tibble(
  name = c("Maple Hall", "Oak Court", "Elm Annex"),
  rent = c("1,250", "980", "missing"),
  occupied = c("TRUE", "FALSE", "N-A")
)
```

   a. (12 points) `rent` is a number even when it contains a comma. `occupied` is logical. Treat `"missing"` and `"N-A"` as missing. Print the cleaned table.

   b. (8 points) With one `across()`, report how many missing values are in each column of the cleaned table.